

Fatemeh azam

Tulsa, OK | Fatemeh-azam@utulsa.edu | 4803599428 | [\[LinkedIn\]](#) [\[Portfolio\]](#) [\[GitHub\]](#)

Software engineer with experience designing and delivering backend and full-stack systems for data-driven, real-time applications, with a focus on cybersecurity, automation, and reliable production systems. Strong background in Python-based services, RESTful APIs, databases, distributed systems, and real-time data pipelines, with experience translating ambiguous technical problems into practical, production-ready solutions. Comfortable working across software, research, hardware, and deployment teams, rapidly prototyping solutions, analyzing system behavior, and communicating technical results to diverse stakeholders. Authorized to work permanently in the U.S. — no employer sponsorship required.

Core Skills

Backend & API Engineering: Python, RESTful API design, Microservices concepts, authentication, validation, performance tuning

Full-Stack Development: React, Node.js, Socket.io, HTML5/CSS3, responsive UI development

Data & Systems: MongoDB, SQL (schema design, indexing, query optimization), SQL Server, real-time data pipelines

DevOps & CI/CD: Git, GitHub Actions, Azure DevOps, unit/integration/API testing

Reliability & Delivery: Production system monitoring, cross-functional delivery, mentoring

Cybersecurity: Security-focused testing, threat/anomaly detection, telemetry analysis, secure-by-design development

Experience

Software Engineer — Oklahoma Cyber Innovation Institute, University of Tulsa

Tulsa, OK | Oct 2024 – Present

- Developed Python-based backend services and RESTful APIs powering interactive, data-driven production applications, integrated with React/Node.js front-ends
- Designed and integrated MongoDB and SQL data pipelines across a distributed multi-node sensor network, supporting real-time analytics, reporting, and reliable system behavior
- Collaborated with cross-functional research, hardware, and deployment teams to translate evolving, ambiguous requirements into maintainable, production-ready solutions
- Analyzed system metrics and performance data to drive reliability improvements, supporting live production deployments and public technical demonstrations
- Partnered with security-focused efforts to test systems against real-world failure and attack scenarios, building secure-by-design practices into development early
- Mentored junior contributors through code review and technical guidance, and presented deployed systems to technical and non-technical audiences at statewide innovation showcases

Project / Event Coordinator — University of California San Diego

San Diego, CA | Nov 2022 – Mar 2024

- Led cross-functional teams and developed streamlined project planning processes, increasing delivery efficiency
- Recognized by supervisor for contributing innovative process strategies

Software Engineer — IRISA Co.

Isfahan, Iran | 2010 – 2016

- Built high-throughput backend modules for log ingestion and processing in Python and C++ using multithreading/multiprocessing
- Designed data pipelines and structured storage workflows supporting real-time analysis at scale
- Developed and integrated anomaly detection and analytics components into production environments, focused on performance, accuracy, and reliability
- Optimized queries and processing for large volumes of system/network events

Education & Publications

M.S. in Software Engineering — University of Isfahan, Iran | 2008–2011

B.S. in Software Engineering — University of Isfahan, Iran | 2003–2008

Accepted paper, "Authority Contention in Micro-UAV Command Stacks" — IEEE CARS 2026 (Autonomous Systems / CAE-R track)